

A Scalable Partial-Information Naming Game

Research plan for lifting a four-agent hidden-profile benchmark to population-scale collective reasoning

Technical notation with plain-language definitions and a worked example

Core idea. Construct tasks in which no single agent can identify the correct answer from its private information, while the union of the population's information is sufficient. Scale the number of agents independently of the number of semantic evidence types, keep communication fully mixed, and measure not only final accuracy but also information diffusion, convention formation, and premature consensus.

1. Scientific objective

The benchmark should preserve the central hidden-profile property: isolated agents are under-informed, but the group is collectively informed. The proposed extension turns this into a population process with explicit information types, repeated or factorized evidence, and a naming-game component that tests whether agents can align different descriptions of the same latent fact.

- **Epistemic requirement:** the correct answer is recoverable only after information is pooled.
- **Population requirement:** the number of agents can grow without requiring one uniquely hand-written clue per agent.
- **Linguistic requirement:** agents may use different surface forms for the same latent evidence and must establish shared conventions.
- **Dynamical requirement:** speaker selection and belief updates define an explicit stochastic process.

2. Scalable construction and complete notation

A task instance contains a finite set of candidate answers, a collection of atomic evidence items, and a population of agents. The notation below separates the semantic complexity of the task from the population size.

Symbol	Technical meaning	Plain-language interpretation
N	Number of agents in the population.	N=20 means that twenty language-model agents participate.
K	Number of candidate answers.	For a multiple-choice task with answers A, B, C, and D, K=4.
M	Number of atomic evidence items.	An atomic item is one minimal fact, such as “Facility Q has certification X.”
C	Number of semantic evidence types.	Several agents may carry evidence of the same type; therefore C need

		not equal N.
i	Index of an individual agent, i in $\{1, \dots, N\}$.	Agent 7 is represented by $i=7$.
e_m	The m -th atomic evidence item, m in $\{1, \dots, M\}$.	e_3 might state that a particular regulation is mandatory.
I_i	The private information set initially given to agent i .	I_i is not a mysterious quantity: it is simply the list of facts visible to agent i before communication.
Y	Random variable representing the unknown correct answer.	Before all evidence is assembled, different answers remain possible.
y^*	The actual correct answer for the task instance.	If answer B is correct, then $y^*=B$.
F	The task-solving rule that maps a collection of evidence to the answer supported by that evidence.	F is not necessarily a learned neural network. It is a mathematical placeholder for the logical or procedural rule of the task.
$\tau(i)$	Evidence type assigned to agent i .	$\tau(i)=2$ means that agent i carries type-2 evidence.
p_c	Population fraction assigned to evidence type c .	$p_2=0.25$ means that 25% of agents carry type-2 evidence.
m_t	Public message broadcast at communication step t .	m_5 is the fifth utterance in the group transcript.
S_i^t	Internal state of agent i after t communication events.	It may contain beliefs, remembered evidence, accepted claims, and lexical conventions.
$q_i^t(y)$	Agent i 's reported probability or confidence for answer y at time t .	For example, $q_i^t(B)=0.7$ means 70% confidence in B.

3. What I_i and F mean

3.1 Private information I_i

For every agent i , the symbol I_i denotes the set of facts initially visible to that agent. If agent 1 receives two facts, e_2 and e_7 , then:

$$I_1 = \{e_2, e_7\}.$$

In simple terms: agent 1 starts with exactly those two pieces of information and does not initially know the remaining facts.

3.2 The solution map F

The symbol F denotes the rule that turns a set of available facts into the answer justified by those facts. It abstracts away the particular task logic. Depending on the benchmark, F could mean applying logical

constraints, reconstructing a graph, evaluating a Boolean circuit, comparing candidates, or executing another deterministic reasoning procedure.

$$F(E) = \text{answer implied by the evidence set } E.$$

The input E may be one agent's private evidence, the public transcript, or the union of all evidence in the population. F is therefore best understood as “solve the task using exactly this evidence.”

3.3 Hidden-profile condition

The benchmark should be generated so that the complete information is sufficient:

$$y^* = F(I_1 \text{ union } I_2 \text{ union } \dots \text{ union } I_N).$$

Plain meaning: after all private facts are pooled, the correct answer y^* can be determined.

At the same time, each isolated agent should remain uncertain:

$$H(Y | I_i) > 0 \text{ for every agent } i.$$

Here $H(Y | I_i)$ is conditional entropy. It is positive when agent i 's private information does not uniquely determine the answer. A simpler implementation-level requirement is that at least two candidate answers remain consistent with I_i .

4. Worked example

Suppose the task asks which candidate is eligible for a position. The answer set is $\{A, B, C, D\}$, so $K=4$. The correct answer is B , but no agent receives the complete argument.

Evidence atom	Content
e1	Candidate B worked at Facility Q.
e2	Facility Q uses protocol X.
e3	Protocol X satisfies requirement R.
e4	Requirement R is mandatory for eligibility.

With four agents, one possible assignment is:

$$I_1=\{e1\}, \quad I_2=\{e2\}, \quad I_3=\{e3\}, \quad I_4=\{e4\}.$$

No individual fact proves eligibility. The complete chain does:

$$B \rightarrow \text{Facility Q} \rightarrow \text{protocol X} \rightarrow \text{requirement R} \rightarrow \text{eligible}.$$

Therefore:

$$F(\{e1,e2,e3,e4\}) = B.$$

This example also shows why F is useful: the same symbol can represent the task's reasoning rule without repeatedly writing the entire logical chain.

5. Scaling beyond four agents

The key design move is to keep the number of evidence types C independent of the number of agents N . A task may have four semantic evidence types but twenty or forty agents.

5.1 Exact replication

Each original evidence type is copied to several agents. For $N=20$ and $C=4$, five agents may receive each type. This tests redundancy, social amplification, and whether frequently repeated evidence is treated as more important than rare evidence.

$$I_i = \{e_{\{\tau(i)\}}\}.$$

Plain meaning: agent i receives the evidence atom associated with its assigned type $\tau(i)$.

5.2 Noisy or paraphrased replication

Agents of the same evidence type receive different linguistic realizations of the same latent fact. Let z_c denote the underlying fact for type c . Then each observed sentence x_i is generated from that latent fact:

$$x_i \sim p(x \mid z_{\{\tau(i)\}}).$$

Plain meaning: two agents may know the same underlying thing but describe it differently. The population must recognize that the statements refer to the same evidence.

5.3 Factorized evidence

A stronger construction divides one semantic fact into several complementary subfacts. Agents of type c collectively reconstruct a latent proposition z_c :

$$z_c = g_c(\text{union of } I_i \text{ over agents with } \tau(i)=c).$$

The function g_c is the local composition rule for evidence type c . After each type-level proposition is reconstructed, the global task rule combines them:

$$y^* = F(z_1, \dots, z_C).$$

This produces two levels of collective reasoning: information must first be integrated within each evidence type and then across evidence types.

6. Naming-game component

The naming component should support reasoning rather than replace it. Agents may initially use different expressions for the same latent evidence object z_c . Through interaction, they develop a shared convention $w_c \leftrightarrow z_c$.

Example: “the certification clue,” “the Facility-Q fact,” and “the compliance chain” may all refer to the same latent object. Successful coordination requires both:

- **Lexical alignment:** agents agree on what an expression refers to.
- **Epistemic integration:** agents correctly combine the aligned evidence to infer y^* .

This separation is scientifically important because a population may converge on terminology while converging on the wrong answer, or solve the task without stabilizing a shared vocabulary.

7. Communication dynamics under full mixing

The first version should use a complete communication graph so that network topology does not confound the study. Every public message is visible to every agent, but only one agent speaks at each microscopic step.

$$s_t \sim \text{Uniform}(\{1, \dots, N\}).$$

The variable s_t is the speaker chosen at time t . Its message m_t is broadcast to the whole population. Each listening agent updates its internal state:

$$S_i^{t+1} = U_i(S_i^t, m_t), \quad \text{for } i \neq s_t.$$

The update rule U_i is whatever process the agent uses to revise beliefs, store evidence, and adjust its vocabulary after reading the message.

To compare different population sizes fairly, define normalized time as:

$$\tau = t / N.$$

One unit of τ corresponds to N speaking events, so each agent has spoken once on average. This is analogous to one Monte Carlo sweep in interacting-particle systems.

8. First implementation program

Stage	Implementation
Task extraction	Select approximately ten clean hidden-profile instances. Represent shared facts, private facts, candidate answers, the correct answer, and a complete-information explanation.
Evidence annotation	Split each task into atomic evidence items e_m and label how each item supports, contradicts, or enables an inference.
Population lifting	Create populations with N in $\{4, 8, 16, 32\}$. Begin with exact replication, then add paraphrased and factorized variants.
Random broadcast process	At each step choose one speaker uniformly, generate one bounded message, and append it to a transcript visible to all agents.
Repeated measurement	At predefined normalized times τ , ask each agent for its answer, confidence, accepted evidence, and vocabulary mappings.
Final evaluation	Compare the population against isolated agents, a full-information single agent, the original four-agent protocol, and an oracle evidence aggregator.

9. Experimental factors

Population sizes: N in $\{4, 8, 16, 32\}$.

Evidence constructions: exact replication, paraphrased replication, and factorized evidence.

Communication budgets: τ in $\{1, 2, 4, 8\}$ normalized sweeps.

Population compositions: balanced types, rare critical evidence, and misleading majority conditions.

Critical-minority parameter: p_{critical} in $\{1/2, 1/4, 1/8, 1/16\}$, where p_{critical} is the fraction of agents possessing the decisive evidence type.

10. Metrics

Metric	Technical definition	Interpretation
Information coverage $C_{\text{info}}(t)$	Fraction of critical evidence atoms that have appeared publicly by time t .	Was the necessary information ever placed into the discussion?
Population penetration $P_e(t)$	Fraction of agents whose current state contains evidence atom e .	Did a fact spread through the population rather than merely being mentioned once?
Answer accuracy $Q(t)$	Fraction of agents currently selecting the correct answer y^* .	How much of the population is correct?
Lexical alignment $A_c(t)$	Degree to which agents use a common expression for latent evidence type c .	Have agents developed a shared name for the same thing?
Consensus time T_{con}	First time at which all or a specified fraction of agents choose the same answer.	How quickly does the group settle?
Premature consensus	Consensus occurs before sufficient critical evidence has entered or propagated through the population.	Did the population agree too early?

11. Recommended first study

The most defensible first experiment is a Typed Replicated Hidden Profile. Use ten benchmark tasks, interpret the original private clues as evidence types, lift each task to several population sizes, paraphrase repeated clues to avoid literal clones, and retain fully mixed random broadcast communication.

Primary research question. Can a fully mixed population of language agents discover and stabilize the semantic conventions needed to integrate distributed evidence, and how does this capability scale with population size, redundancy, and the rarity of critical information?

12. Minimal pseudocode

Input: task instance, population size N , evidence allocation, communication budget T

- Construct agents $i=1,\dots,N$ and assign each private information set I_i .
- Record every agent's initial answer and confidence.
- For $t=1,\dots,T$:
 - Sample speaker s_t uniformly from the population.
 - Generate public message m_t from the speaker's private evidence and transcript.
 - Broadcast m_t to all agents.
 - Update each listener state S_i^t .
 - At measurement times, record answers, evidence states, and lexical mappings.
- Compute final accuracy, information diffusion, lexical alignment, and consensus metrics.